



ECEN 5793

Digital Image Processing

Nate Lannan

Agenda

Introduce general information about this course

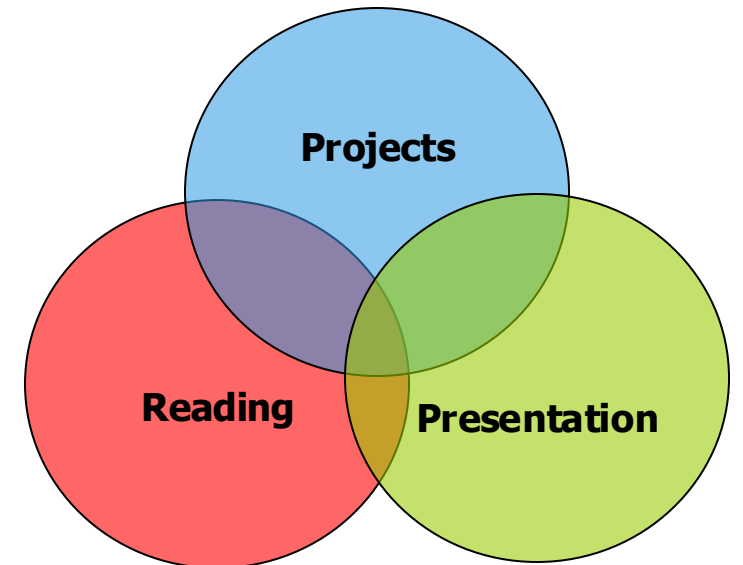
Introduce Helpful Materials

Introduce the main projects in this class

Course Expectations

Helpful practices

- Attending the class
- Attending Office Hours (ES 267, WR 1:00-2:30PM)
- Doing projects independently and promptly
- Reading supplementary materials

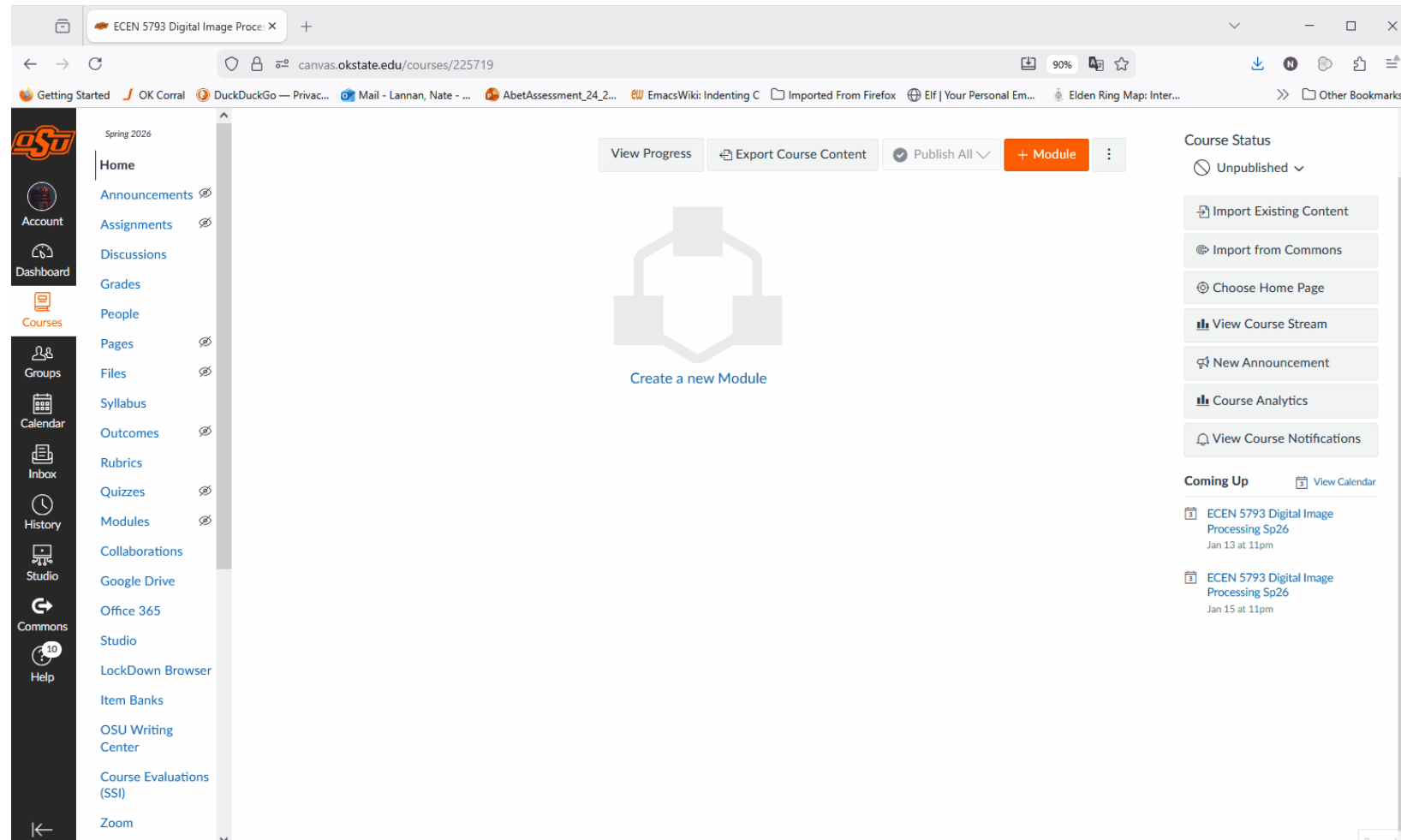


Zoom Access

In-person class attendance is **strongly encouraged**.

- Online attendance via ZOOM (937 4098 0202, Passcode: Image)
- Offline video watching will also be available

All classes will be recorded, and lecture videos will be made available online in Canvas for review.



Grading Policy

The final grade is based on

- All six computer projects (85%)
- One student presentation (15%)

The letter grade is based on

- ≥ 90 : A
- 80-89: B
- 70-79: C
- 60-69: D
- < 60 : F

Suggested Reading and Prerequisites

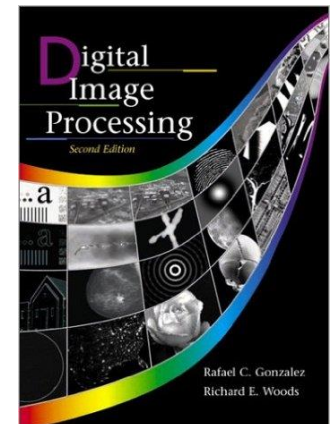
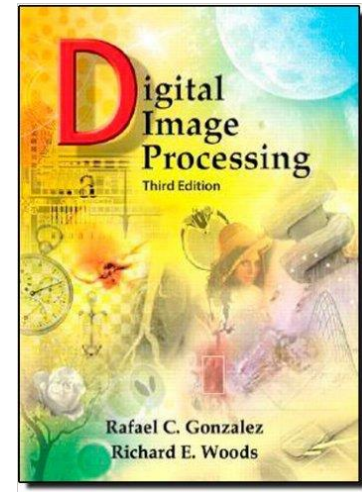
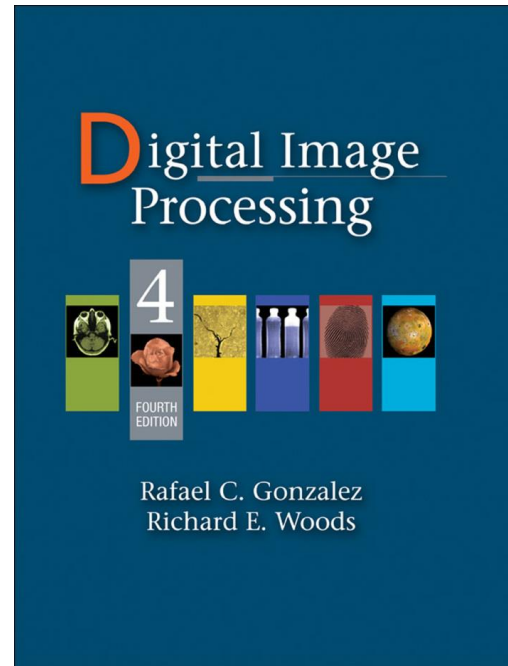
Main Reference

Digital Image Processing 4e

Pearson

Gonzales and Woods

ISBN: 978-0-13-335672-4



Prerequisites

- ECEN 4763/5763 Digital Signal Processing
- Python background is plus

Suggested Reading and Prerequisites

Supplementary References

Deep Learning for Vision Systems

Manning

Mohamed Elgendy

ISBN: 978-1-61-729619-2

Learning OpenCV 4 Computer Vision with Python 3 3e

Packt

Joseph Howse and Joe Minichino

ISBN: 978-1-78-953161-9

Suggested Reading and Prerequisites

Supplementary References

<https://github.com/kjw0612/awesome-deep-vision#books>

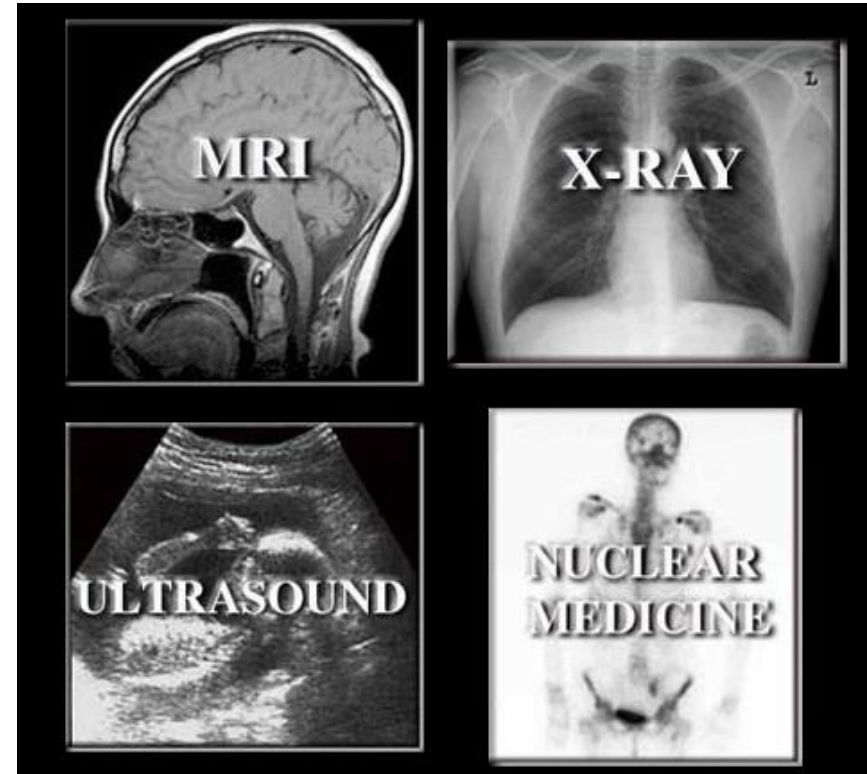
Why do we need images?

One picture is worth more than ten thousand words.

Five senses

- Sight
- Hearing
- Touch
- Taste
- Smell

Often imaging is the interface to computer systems for data that is beyond our sensory perception



Digital Image Processing (DIP)

An **image** is defined a two-dimensional function $f(x, y)$

- x and y are spatial (plane) coordinates
- the amplitude $f(x, y)$ of coordinate (x, y) is called intensity or gray level.

A **digital image** is an image when x , y , and the amplitude values of $f(x, y)$ are all finite, discrete quantities.

- **DIP** refers to processing digital images by means of a digital computer.
- **Pixels** are referred to basic image elements which consist an image.

How about a color image?

$$f(x, y, c) \text{ where } c = 1, 2, 3$$

How about a color video?

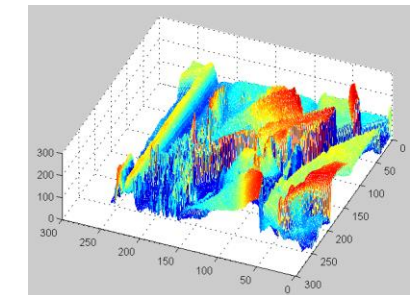
$$f(x, y, c, t)$$



2-bit



10-bit



Class Outline

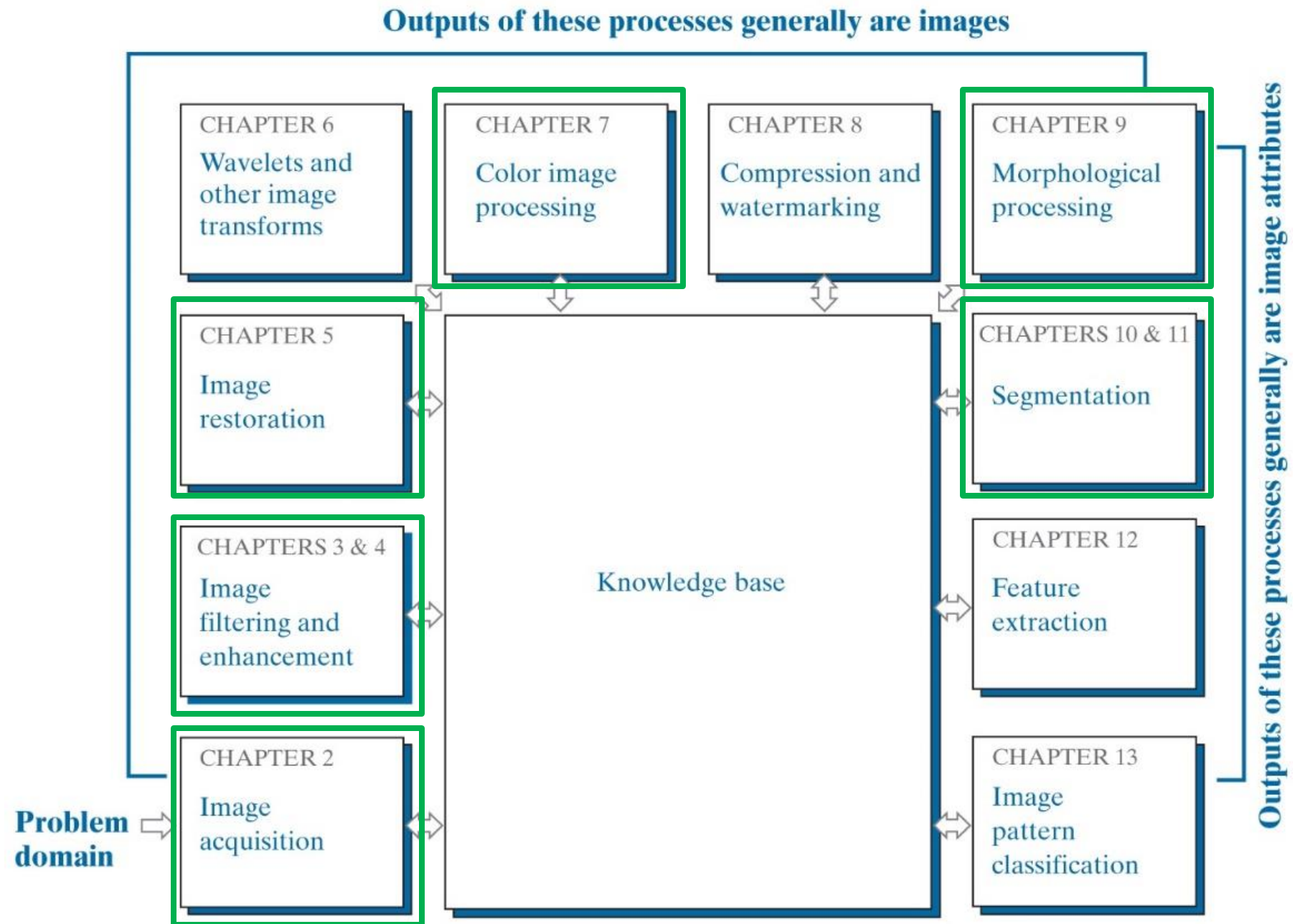


Image Acquisition

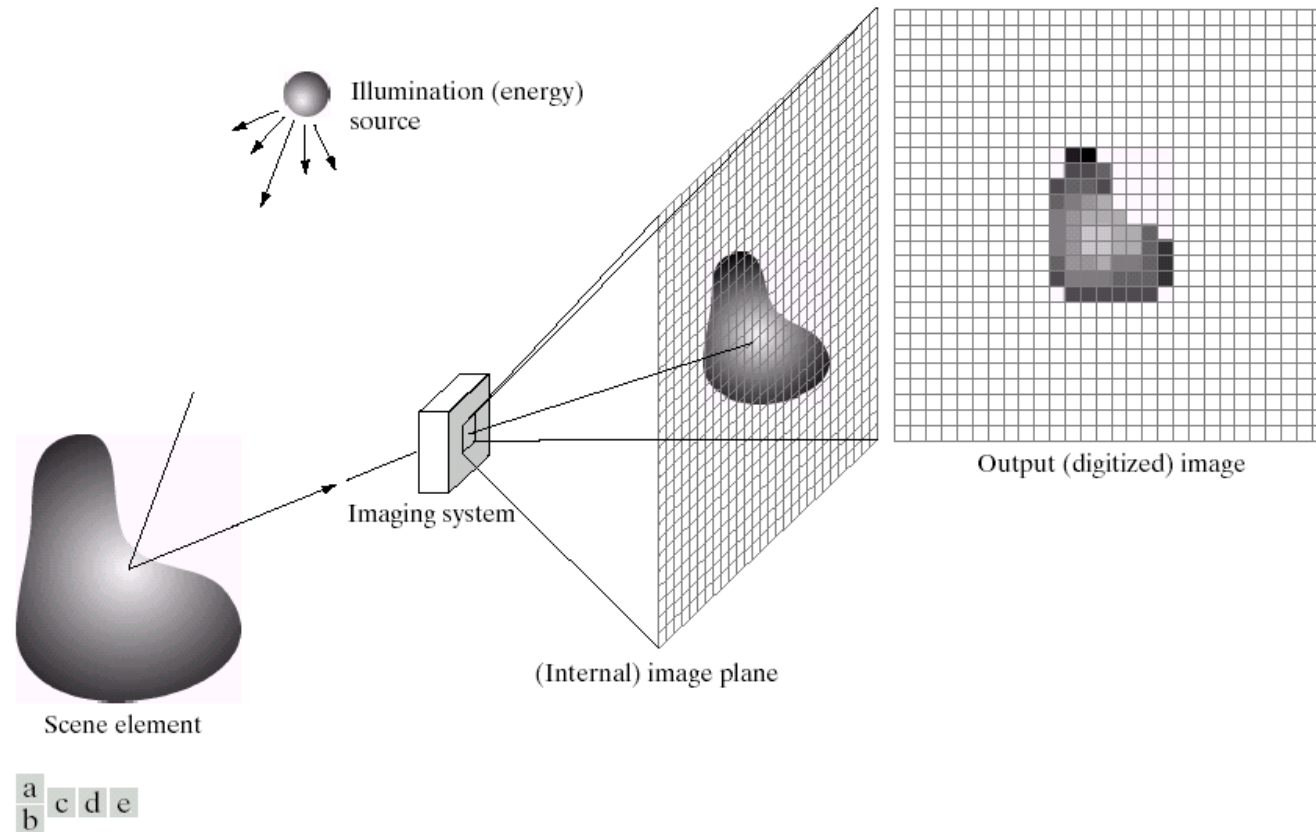
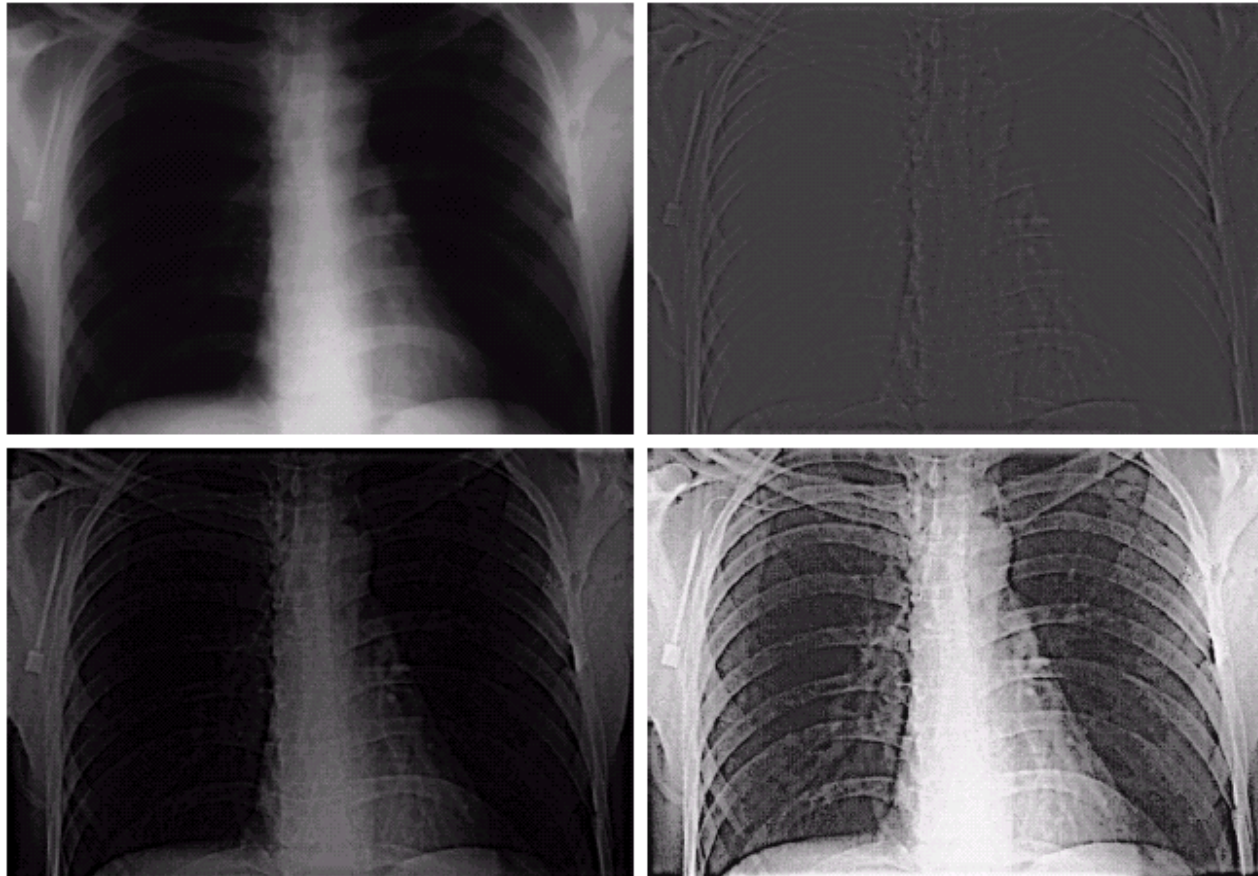


FIGURE 2.15 An example of the digital image acquisition process. (a) Energy (“illumination”) source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

Image Enhancement

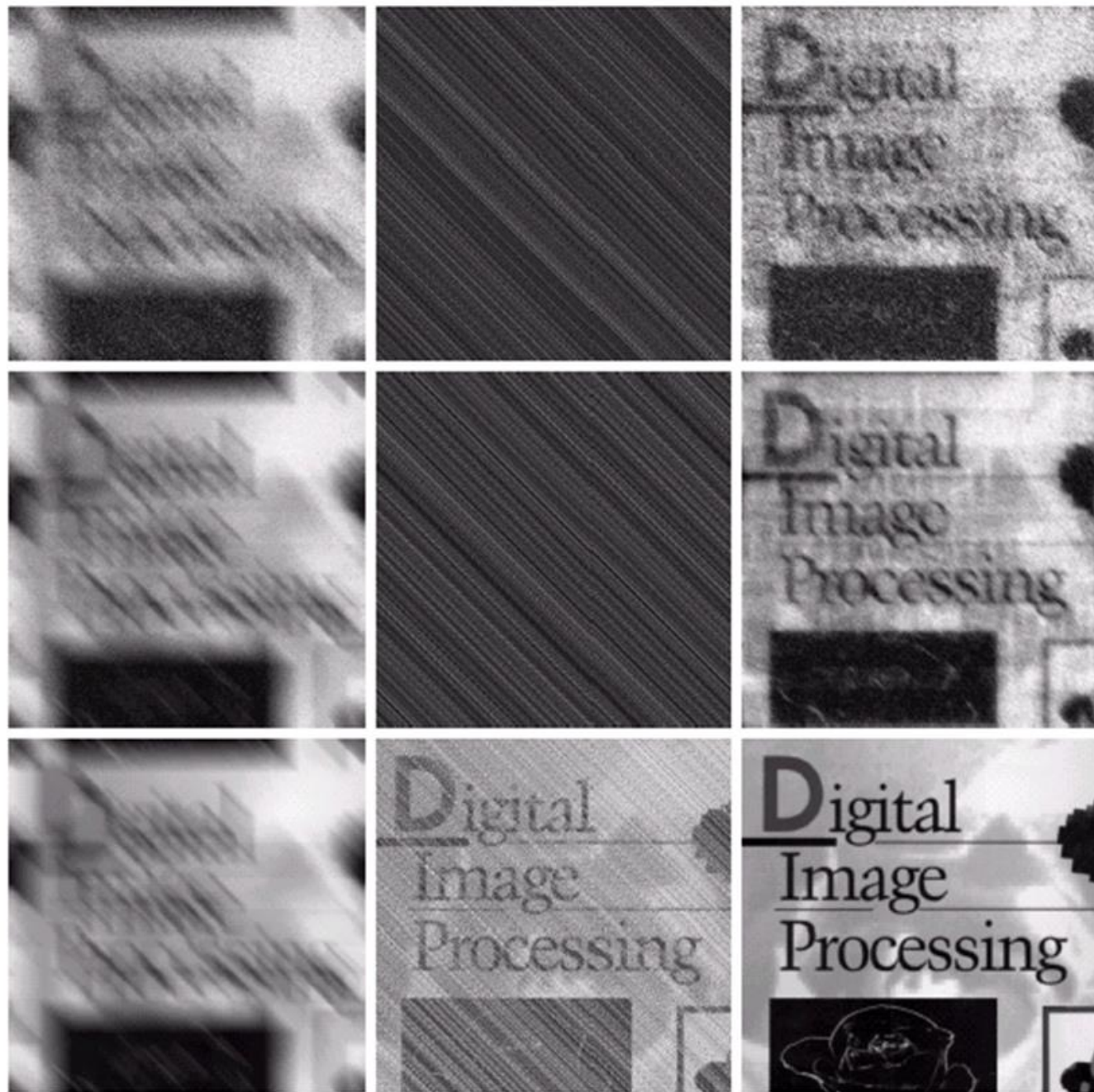


a	b
c	d

FIGURE 4.30

(a) A chest X-ray image. (b) Result of Butterworth highpass filtering. (c) Result of high-frequency emphasis filtering. (d) Result of performing histogram equalization on (c). (Original image courtesy Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School.)

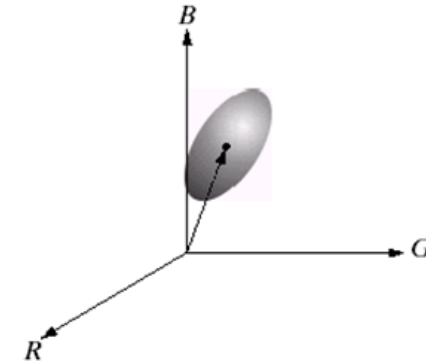
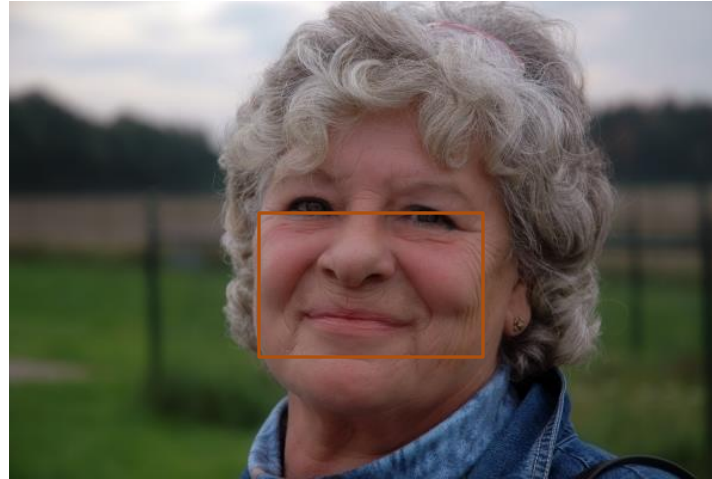
Image Restoration



a b c
d e f
g h i

FIGURE 5.29 (a) Image corrupted by motion blur and additive noise. (b) Result of inverse filtering. (c) Result of Wiener filtering. (d)–(f) Same sequence, but with noise variance one order of magnitude less. (g)–(i) Same sequence, but noise variance reduced by five orders of magnitude from (a). Note in (h) how the deblurred image is quite visible through a “curtain” of noise.

Color Image Processing



Morphological Processing

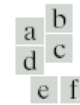
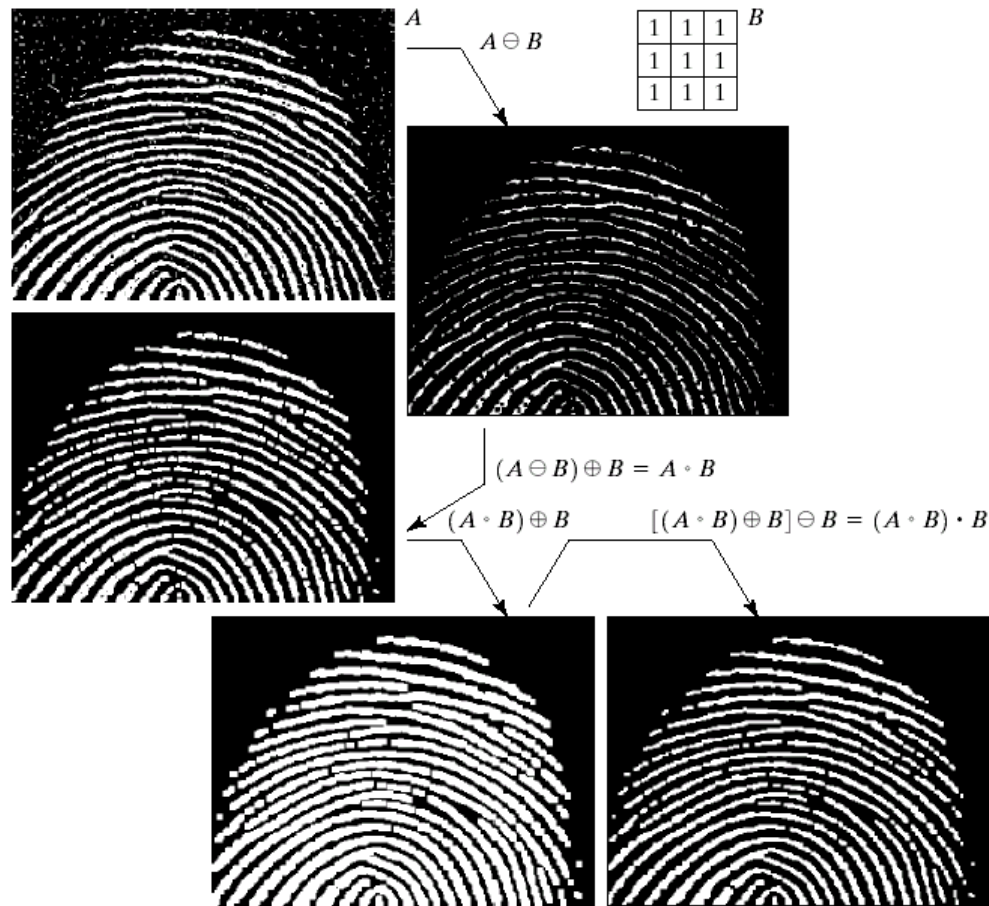
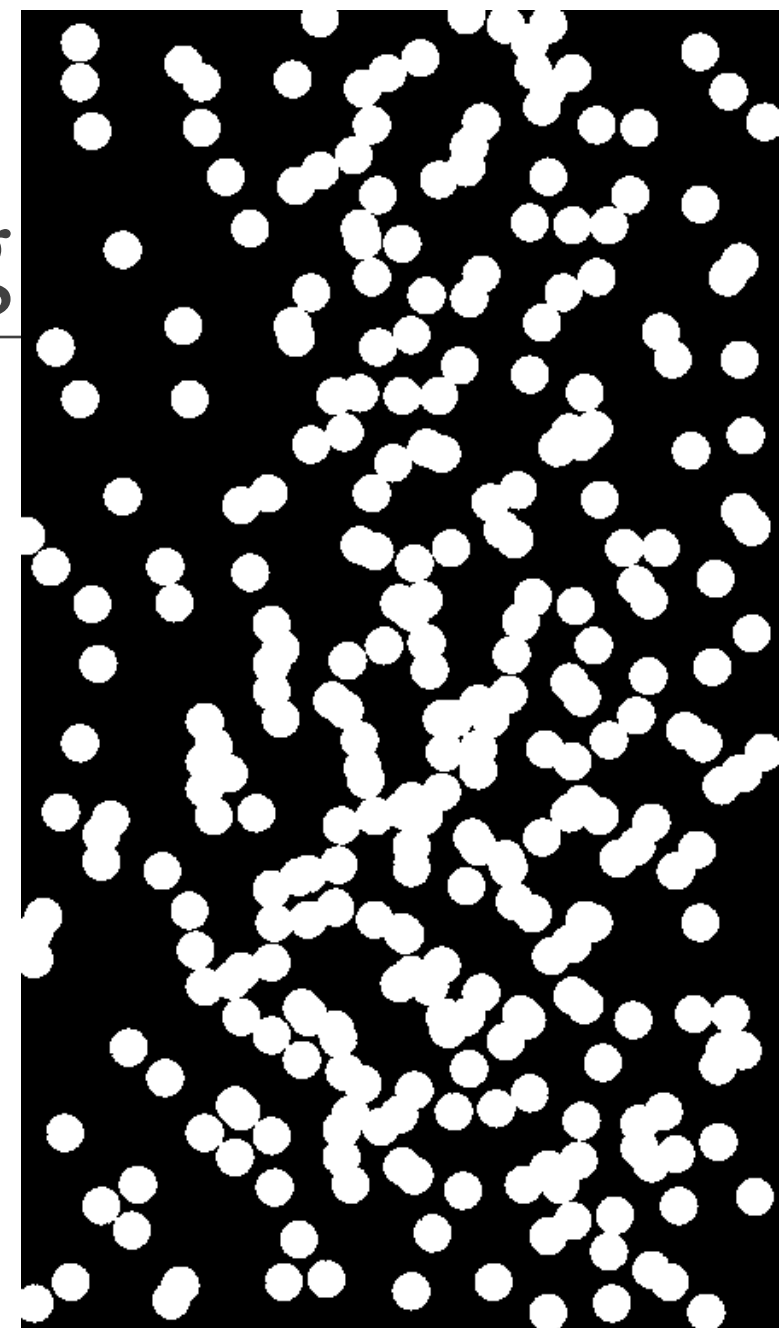
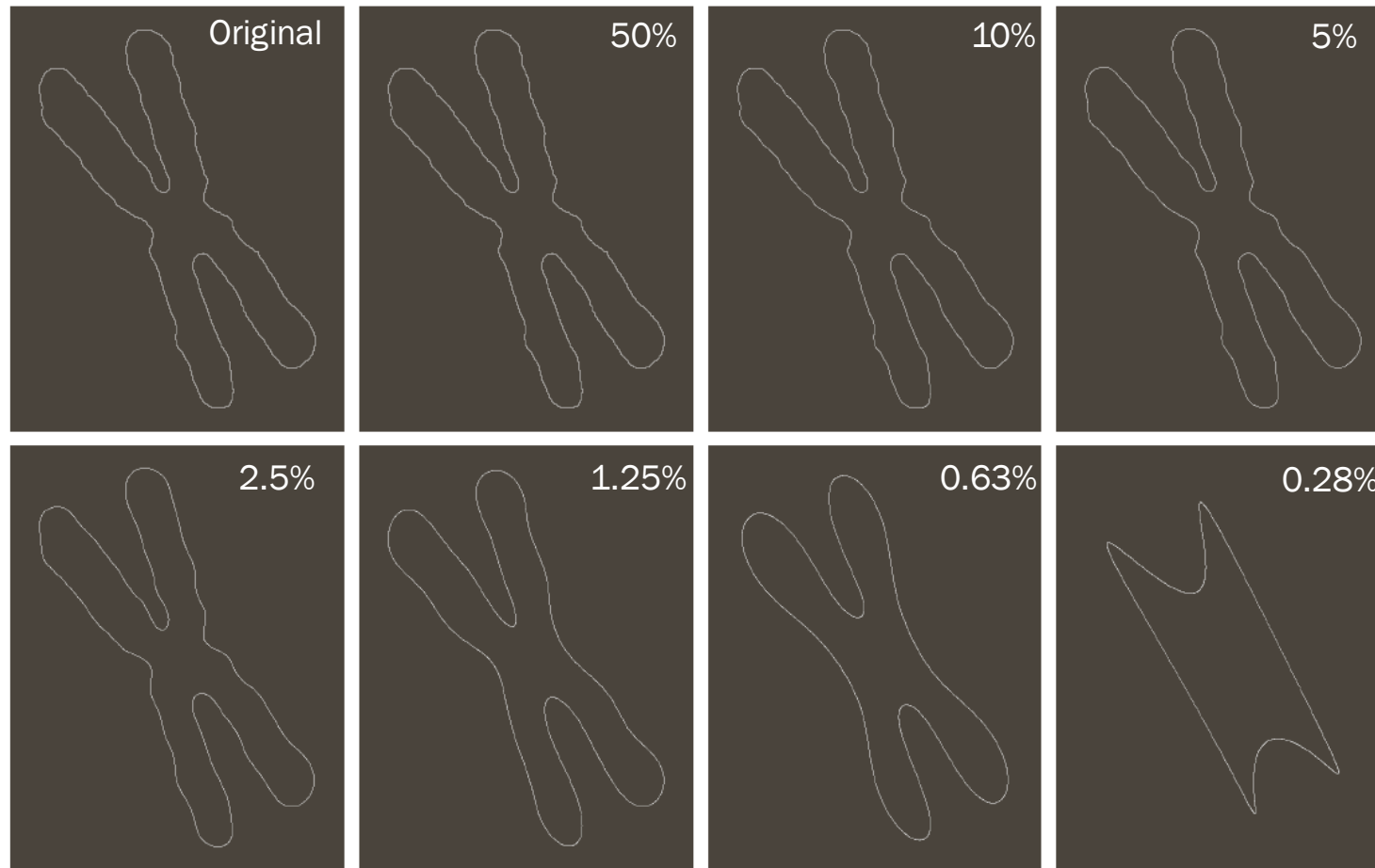


FIGURE 9.11

(a) Noisy image.
 (c) Eroded image.
 (d) Opening of A .
 (e) Dilation of the opening.
 (f) Closing of the opening. (Original image for this example courtesy of the National Institute of Standards and Technology.)



Shape Analysis of Objects



Shape Representation of Objects

